ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Health: The Ultimate Treasure** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** 70	40 + 30 = 70.
3	**(d)** colds are viral and often recover by themselves	Colds go away without them.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Antibodies prepare the body.
5	**(b)** 75%	15000/20000 x 100 = 75%.
6	**(d)** boiling milk	Not related to vector.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Breeding sites are removed.
8	**(a)** The pathogen can spread through unvaccinated people	Community protection weakens.
9	**(d)** A is false but R is true.	A is false because health is complete well-being; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes how the mosquito becomes infected, and A states the route to a new person, so R does not directly explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Unvaccinated = 2000 = 25% (1); Fewer susceptible persons means less spread (1).	1+1
12	They cannot be cured completely (1); Treatment and lifestyle changes keep them in control (1).	1+1
13	Prevent contamination of water (1); Reduce fly and mosquito breeding (1).	1+1
14	Flu is viral so antibiotics do not work (1); Rest and fluids help the immune system (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Contaminated drinking water (1); Unhygienic food (1); Preventive: boiled water, hygiene, treat water tank (1).	1+1+1
16	Goitre: iodised salt (1); Rickets: milk, sunlight (1); Anaemia: leafy vegetables (1).	1+1+1
17	Clean water and mess hygiene (1); Regular medical check-ups (1); Mosquito control and cleanliness (1).	1+1+1
	OR: Vaccine prevents infection; does not cure (1); Medicine treats symptoms (1); Measles has no antibiotic cure as it is viral (1).	1+1+1
18	Individual: repellent, long sleeves (1); Family: nets, no stagnant water (1); Community: fogging, drain cleaning (1).	1+1+1
	OR: Vitamin and fibre deficiency (1); Poor digestion, weak immunity (1); Include seasonal fruit and vegetables (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Malaria (1). (ii) Female Anopheles mosquito (1). (iii) A protozoan (Plasmodium) parasite (1). (iv) Do not let water stagnate (empty coolers, tyres, pots weekly); use mosquito nets/repellents; spray insecticide or oil on ponds (any two) (2).	1+1+1+2
20	First dose: slow, small rise (1); Second dose: faster, higher rise (1); Memory cells remain (1); Levels fall later (1); Booster restores (1).	1+1+1+1+1
	OR: Personal habits: hygiene, diet (1); Family: nutrition, emotional support (1); Community: water, sanitation, healthcare (1); Policies and awareness (1); Health depends on all three (1).	1+1+1+1+1

Mistakes to avoid:

- Defining health as only the absence of disease; the chapter includes mental and social health.
- Writing that all diseases spread from person to person; diabetes and cancer do not.

- Mixing carriers: malaria is spread by Anopheles, dengue by Aedes.
- Saying antibiotics cure viral diseases like influenza or dengue.
- Confusing a vaccine (prevents) with a medicine (treats).
- Naming only the disease and forgetting its cause, carrier and prevention when asked for a full answer.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Health: The Ultimate Treasure** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** typhoid	Faecal contamination.
2	**(b)** Aedes breeds in small collections of clean water	Coolers and pots.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Heat kills microbes.
4	**(a)** tuberculosis	A chronic bacterial infection.
5	**(d)** difference in immunity	Immune strength varies.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Vitamin C is the missing nutrient.
7	**(d)** sharing a handkerchief	Sharing spreads germs.
8	**(b)** iodine - goitre	Correct pairing.
9	**(d)** A is false but R is true.	Diabetes is non-communicable, so A is false; R is the correct definition of communicable diseases.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The immune memory created by the vaccine explains the protection.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The first dose primes the immune system (1); The second dose strengthens memory (1).	1+1
12	Food gives energy, proteins, fibre also (1); Tablets give only some nutrients (1).	1+1
13	To prevent entry of germs (1); Dirty wounds may cause tetanus or other infections (1).	1+1
14	Mosquitoes inject pathogens when they bite (1); Houseflies carry germs on their legs to food (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hands carry germs from many surfaces (1); Washing removes them (1); Gloves can also get contaminated (1).	1+1+1
16	The bacteria multiply slowly (1); Short treatment may leave them alive (1); Drug resistance can develop (1).	1+1+1
17	To avoid resistance (1); Dose and duration depend on the infection (1); Side effects are avoided (1).	1+1+1
	OR: Droplets carry pathogens (1); Others inhale droplets (1); Cover the mouth; ventilate the room (1).	1+1+1
18	Each dose strengthens immunity (1); More children are covered (1); Eradicates the virus (1).	1+1+1
	OR: Stress hormones weaken defence (1); Poor sleep and eating also matter (1); Exercise and talking help (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Non-communicable (1). (ii) Chickenpox or measles (1). (iii) It stimulates the immune system to produce defence against the germ (1). (iv) Chickenpox is caused by a virus that passes from person to person; diabetes is not caused by a germ, so there is nothing to pass on (2).	1+1+1+2
20	Causes: pathogens vs lifestyle/genes (1); Spread: yes vs no (1); Prevention: hygiene, vaccines vs diet, exercise (1); Examples: cholera, TB vs diabetes, heart disease (1); Treatment: antibiotics vs long care (1).	1+1+1+1+1
	OR: Total with at least one = 12 + 8 - 4 (1); = 16 (1); Neither = 40 - 16 = 24 (1); Measure: remove stagnant water (1); Awareness drive (1).	1+1+1+1+1

Mistakes to avoid:

- Defining health as only the absence of disease; the chapter includes mental and social health.
- Writing that all diseases spread from person to person; diabetes and cancer do not.
- Mixing carriers: malaria is spread by Anopheles, dengue by Aedes.
- Saying antibiotics cure viral diseases like influenza or dengue.

- Confusing a vaccine (prevents) with a medicine (treats).
- Naming only the disease and forgetting its cause, carrier and prevention when asked for a full answer.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Health: The Ultimate Treasure** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** faecal-oral transmission	Hands carry faecal germs.
2	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but R explains why control is needed, not how spread occurs.
3	**(c)** restores lost water and salts	Treats dehydration.
4	**(c)** is caused by lack of a nutrient	Not by pathogens.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(c)** malaria	Malaria is spread by the Anopheles mosquito.
7	**(a)** HIV and hepatitis	To keep the blood safe.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Memory cells.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Removing germs before they enter the mouth reduces infection.
10	**(c)** A is true but R is false.	A is true; R is false: preventing breeding by removing stagnant water is the measure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Breeding places of mosquitoes are common (1); Poor sanitation and standing water (1).	1+1
12	Boiling kills pathogens at home (1); Chlorine disinfects large volumes (1).	1+1
13	Damages the user's lungs (1); Passive smoking affects others (1).	1+1
14	Strengthens heart, muscles (1); Improves mood and sleep (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fever can be due to infection (1); Medicine reduces fever, not always the cause (1); Doctor's test is needed (1).	1+1+1
16	More illness (1); Loss of work days (1); Medical costs rise (1).	1+1+1
17	Prevention: vaccines, hygiene (1); Cure: antibiotics, treatment (1); Prevention is better than cure (1).	1+1+1
	OR: Memory cells already exist (1); Antibodies are made quickly (1); Basis of vaccination (1).	1+1+1
18	Carrier has pathogen but no symptoms (1); Patient shows signs and symptoms (1); e.g. typhoid carrier (1).	1+1+1
	OR: When most people are immune (1); The pathogen finds few hosts (1); Spread is reduced for all (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Physical, mental and social (any two) (1). (ii) Mental (and social) health (1). (iii) Balanced diet / sleep / exercise / cleanliness (any one) (1). (iv) Health is complete well-being, not just the absence of disease; emotional and social problems also count (2).	1+1+1+2
20	Vitamin B1 deficiency possible (beriberi) (1); Scurvy due to no vitamin C (1); Anaemia due to low iron (1); Add whole grains, fruits, green vegetables (1); Include pulses (1).	1+1+1+1+1
	OR: Pulse polio campaign (1); Hand washing drives (1); Community clean-up drives (1); Awareness on dengue and malaria (1); Informed people prevent disease (1).	1+1+1+1+1

Mistakes to avoid:

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